



C++ Programming

Intern

Name: Amima Batool

Starting Date: September 01, 2026

Ending Date: September 30, 2026

Project no. : First

Project name: CGPA Calculator

CGPA Calculator

Application

Introduction:

CGPA Calculator is a simple command-line application developed using C++. The program allows the user to enter a student's name, current semester, number of courses, course details, credit hours, and grades. The program then calculates the current semester GPA and overall CGPA.

The application uses classes, objects, vectors, loops, functions, conditional statements, and user input to manage student and course information. It also allows the user to enter the GPA of previous semesters to calculate the overall CGPA.

This project was developed to practice fundamental C++ programming concepts and object-oriented programming..

Tools & Technologies Used:

- VS Code
- C++
- GitHub

Objectives of the Project:

The main objectives of the CGPA Calculator are:

- To develop a simple CGPA calculation application using C++.
- To practice object-oriented programming concepts.
- To understand and implement classes and objects.
- To learn how to use constructors.
- To practice working with vectors.
- To understand the use of functions inside classes.
- To practice loops and conditional statements.
- To take and process user input.
- To calculate semester GPA using credit hours and grade points.
- To calculate overall CGPA using previous semester GPAs and current semester GPA.
- To gain hands-on experience in developing a command-line application.

Features:

Following are the features of the CGPA Calculator:

1. Allows the user to enter the student's name.
2. Allows the user to enter the current semester.
3. Allows the user to enter the number of courses.
4. Takes the name of each course as input.

5. Takes credit hours for each course.
6. Takes the grade/grade point for each course.
7. Stores course information using objects.
8. Stores multiple subjects using a vector.
9. Calculates total credit hours.
10. Calculates total grade points.
11. Calculates the current semester GPA.
12. Allows the user to enter GPA of previous semesters.
13. Calculates the overall CGPA.
14. Displays detailed course information.
15. Displays the semester GPA and CGPA.
16. Provides a simple and easy-to-use command-line interface.

C++ Concepts Used:

Following fundamental C++ Concepts are used in this project:

➤ Classes

```
class subject
{
public:
    string subject_name;
```

```
class student
{
public:
    string name;
    int current_semester;
```

➤ Objects

```
student student(student_name, current_semester);
```

```
student.add_subject(subject_name, credits, Grade);
```

➤ Constructors

```
//constructor
subject(string name, int c, float g)
{
    subject name = name;
```

```
//constructor
student(string n, int sem)
{
    name = n;
```

➤ Encapsulation

```
class subject
{
public:
    string subject_name;
    int credits;
    float Grade;
```

```
class student
{
public:
    string name;
    int current_semester;
    vector<subject> subjects;
```

➤ Member functions

```

vector<subject> subjects;

//constructor
student(string n, int sem)
{
    name = n;
    current_semester = sem;
}

//function to add subject
void add_subject(string sub_name, int c, float g)
{
    subject s(sub_name, c, g);
    subjects.push_back(s); //pushing in vector
}

//function to calculate total credits
float total_credit()
{
    float total_credits = 0;
    for(const subject & s: subjects)

```

- Public data members

```

class subject
{
public:
    string subject_name;
    int credits;
    float Grade;
}

```

```

class student
{
public:
    string name;
    int current_semester;
    vector<subject> subjects;
}

```

- Variables and Data types ("string", "int", "float")

```

class subject
{
public:
    string subject_name;
    int credits;
    float Grade;
}

```

- Vectors

```

int current_semester;
vector<subject> subjects;

```

- "vector::push_back()"

```

subject s(sub_name, c, g);
subjects.push_back(s); //pushing in vector

```

- "for" loops

```

cout << "Subjects: " << endl;
for(const subject & s: subjects)
{
    cout << "Course Name: " << s.subject_name << endl;
    cout << "Credit Hours: " << s.credits << endl;
    cout << "Grade: " << s.Grade << endl;
    cout << "Grade Points: " << s.Grade * s.credits << endl;
    cout << endl;
}

```

- User input using "cin" and Output using "cout"

```

cout << "Enter current semester: ";
cin >> current_semester;

cout << "Enter number of Courses: ";
cin >> name_cou;
cout << endl;

```

- "getline()" and "cin.ignore()"

```
cout<<"Enter name of course "<<i+1<<": ";
cin.ignore();           // To ignore the newline character left in the input buffer
getline(cin , subject_name);
```

- Arithmetic operations

```
total_grade += s.Grade * s.credits;
```

```
cin>>pre_sem_GPA;
total_CGPA += pre_sem_GPA;
```

```
return (total_credits > 0) ? (total_grade / total_credits) : 0;
```

- Header files and Standard namespace

```
#include<iostream>
#include<vector>
using namespace std;
```

Working of the Program:

The execution process of the program is as follows:

1. Two classes, "subject" and "student", are created. The "subject" class stores the name, credit hours, and grade of each course. The "student" class stores the student's name, current semester, and a vector containing the subjects.

```
class subject
{
public:
    string subject_name;
```

```
class student
{
public:
    string name;
    int current_semester;
```

2. The program starts by displaying the title "CGPA Calculator".

```
int main()
{
    cout<<endl;
    cout<<"-----CGPA Calculator-----"<<endl;

    string student_name;
```

3. The user is asked to enter the student's name and current semester. The user also enters the number of courses they have studied in the current semester.

```
string student_name;
int current_semester;
int name_cou;

cout<<"Enter name of Student: ";
getline(cin , student_name);

cout<<"Enter current semester: ";
cin>>current_semester;

cout<<"Enter number of Courses: ";
cin>>name_cou;
cout<<endl;
```

4. A "for" loop runs according to the number of courses entered. For each course, the program asks the user to enter: Course name, Credit hours and Grade/grade point.

```

student student(student_name, current_semester);

for(int i=0; i<name_cou; i++)
{
    string subject_name;
    int credits;
    float Grade;

    cout<<"Course "<<i+1<<": "<<endl;
    cout<<"Enter name of course "<<i+1<<": ";
    cin.ignore(); // To ignore the newline character left in the input buffer
    getline(cin, subject_name);

    cout<<"Enter credits for course '"<<subject_name<<": ";
    cin>>credits;

    cout<<"Enter Grade for course '"<<subject_name<<": ";
    cin>>Grade;
    cout<<endl;

    student.add_subject(subject_name, credits, Grade);
}

```

5. The "add_subject()" function creates a subject object and stores it in the student's "subjects" vector using "push_back()".

```

//function to add subject
void add_subject(string sub_name, int c, float g)
{
    subject s(sub_name, c, g);
    subjects.push_back(s); //pushing in vector
}

```

6. After all course information is entered, the "result()" function displays the student's information and details of all courses.

```

    cout<<endl;
    student.result();
    cout<<endl;
}

```

7. The "total_credit()" function uses a range-based "for" loop to calculate the total credit hours of all courses.

```

//function to calculate total credits
float total_credit()
{
    float total_credits = 0;
    for(const subject & s: subjects)
    {
        total_credits += s.credits;
    }
    return total_credits;
}

```

8. The "total_grades()" function calculates the total grade points by multiplying the grade of each course by its credit hours.

```

//function to calculate total grade points
float total_grades()
{
    float total_grade = 0;
    for(const subject & s: subjects)
    {
        total_grade += s.Grade * s.credits;
    }
    return total_grade;
}

```

9. The "calculate_GPA()" function calculates the current semester GPA.

```
//function to calculate current semester GPA
float calculate_GPA()
{
    float total_credits = total_credit();
    float total_grade = total_grades();

    return (total_credits > 0) ? (total_grade / total_credits) : 0;
}
```

10. The "calculate_CGPA()" function asks the user to enter the GPA of all previous semesters. The previous semester GPAs are added together with the current semester GPA. The overall CGPA is calculated using: $CGPA = \text{Total Semester GPAs} / \text{Total Number of Semesters}$

```
//function to calculate CGPA
float calculate_CGPA()
{
    float total_CGPA = 0;

    for(int i=1; i<current_semester; i++)
    {
        float pre_sem_GPA= 0;
        cout<<"Enter GPA of Semester "<<i<<" : ";
        cin>>pre_sem_GPA;
        total_CGPA += pre_sem_GPA;
    }

    total_CGPA += calculate_GPA();
    int total_semesters = current_semester;
    return total_CGPA/total_semesters;
}
```

11. Finally, the program displays the student's current semester GPA and overall CGPA.

```
//function to display Course details
void result()
{
    cout<<"-----RESULT-----"<<endl;

    cout << "Student Name: " << name << endl;
    cout<< "Current Semester: " << current_semester<<endl;

    cout<<endl;
    cout<<"-----Course Details-----"<<endl;

    cout << "Subjects: " << endl;
    for(const subject & s: subjects)
    {
        cout << "Course Name: " << s.subject_name <<endl;
        cout<< "Credit Hours: " << s.credits <<endl;
        cout<< "Grade: " << s.Grade << endl;
        cout<<"Grade Points: " << s.Grade * s.credits << endl;
        cout<<endl;
    }

    cout<<"-----Current Semester GPA Details-----"<<endl;
    cout<<"Total Credits: " << total_credit() << endl;
    cout<<"Total Grade Points: " << total_grades() << endl;
    cout<<"Semester GPA: " << calculate_GPA() << endl;
    cout<<endl;

    cout<<"-----CGPA Details-----"<<endl;
    float cgpa = calculate_CGPA();
    cout << "CGPA of " << name << " is: " << cgpa;
    cout<<endl;
}
```

Application Flow:

The overall basic workflow of the application is as follow:

Start → Display "CGPA Calculator" → Enter Student Name → Enter Current Semester → Enter Number of Courses → Enter Course Details → Store Course Information → Calculate Total Credits → Calculate Total Grade Points → Calculate Current Semester GPA → Enter Previous Semester GPA → Calculate CGPA → Display Course Details → Display GPA and CGPA → END

Code:

// CGPA Calculator

#include<iostream>

#include<vector>

using namespace std;

class subject

{

public:

string subject_name;

int credits;

float Grade;

//constructor

subject(string name, int c, float g)

{

subject_name = name;

credits = c;

Grade = g;

}

};

class student

{

public:

string name;

int current_semester;

vector<subject> subjects;

//constructor

student(string n, int sem)

{

name = n;

current_semester= sem;

}

```

//function to add subject
void add_subject(string sub_name, int c, float g)
{
    subject s(sub_name, c, g);
    subjects.push_back(s); //pushing in vector
}

//function to calculate total credits
float total_credit()
{
    float total_credits = 0;
    for(const subject & s: subjects)
    {
        total_credits += s.credits;
    }
    return total_credits;
}

//function to calculate total grade points
float total_grades()
{
    float total_grade = 0;
    for(const subject & s: subjects)
    {
        total_grade += s.Grade * s.credits;
    }
    return total_grade;
}

//function to calculate current semester GPA
float calculate_GPA()
{
    float total_credits = total_credit();
    float total_grade = total_grades();

    return (total_credits > 0) ? (total_grade / total_credits) : 0;
}

//function to calculate CGPA
float calculate_CGPA()
{
    float total_CGPA = 0;

    for(int i=1; i<current_semester; i++)
    {
        float pre_sem_GPA= 0;

```

```

        cout<<"Enter GPA of Semester "<<i<<" ": ";
        cin>>pre_sem_GPA;
        total_CGPA += pre_sem_GPA;
    }

    total_CGPA += calculate_GPA();
    int total_semesters = current_semester;
    return total_CGPA/total_semesters;
}

//function to display Course details
void result()
{
    cout<<"=====--RESULT--===== "<<endl;

    cout << "Student Name: " << name << endl;
    cout<< "Current Semester: " << current_semester<<endl;

    cout<<endl;
    cout<<"-----Course Details-----" <<endl;

    cout << "Subjects: " << endl;
    for(const subject & s: subjects)
    {
        cout << "Course Name: " << s.subject_name <<endl;
        cout<< "Credit Hours: " << s.credits <<endl;
        cout<< "Grade: " << s.Grade << endl;
        cout<<"Grade Points: " << s.Grade * s.credits << endl;
        cout<<endl;
    }

    cout<<"-----Current Semester GPA Details-----" <<endl;
    cout<<"Total Credits: " << total_credit() << endl;
    cout<<"Total Grade Points: " << total_grades() << endl;
    cout<<"Semester GPA: " << calculate_GPA() << endl;
    cout<<endl;

    cout<<"-----CGPA Details-----" <<endl;
    float cgpa = calculate_CGPA();
    cout << "CGPA of " << name << " is: " << cgpa;
    cout<<endl;
}

};

```

```

int main()
{
    cout<<endl;
    cout<<"---==---CGPA Calculator---==---"<<endl;

    string student_name;
    int current_semester;
    int name_cou;

    cout<<"Enter name of Student: ";
    getline(cin , student_name);

    cout<<"Enter current semester: ";
    cin>>current_semester;

    cout<<"Enter number of Courses: ";
    cin>>name_cou;
    cout<<endl;

    student student(student_name, current_semester);

    for(int i=0; i<name_cou; i++)
    {
        string subject_name;
        int credits;
        float Grade;

        cout<<"Course "<<i+1<<": "<<endl;
        cout<<"Enter name of course "<<i+1<<": ";
        cin.ignore();          // To ignore the newline character left in the input buffer
        getline(cin , subject_name);

        cout<<"Enter credits for course '"<<subject_name<<"' ";
        cin>>credits;

        cout<<"Enter Grade for course '"<<subject_name<<"' ";
        cin>>Grade;
        cout<<endl;

        student.add_subject(subject_name, credits, Grade);
    }
    cout<<endl;
    student.result();
    cout<<endl;
}

```

Sample Output:

```
PS C:\Users\amima\OneDrive\Desktop\VS CODE folder> cd "c:\Users\amima\OneDrive\Desktop\VS CODE folder"
PA-Calculator }

-----CGPA Calculator-----
Enter name of Student: Amima
Enter current semester: 4
Enter number of Courses: 5

Course 1:
Enter name of course 1: DSA
Enter credits for course 'DSA': 4
Enter Grade for course 'DSA': 3.95

Course 2:
Enter name of course 2: Signals and Systems
Enter credits for course 'Signals and Systems': 4
Enter Grade for course 'Signals and Systems': 3.98

Course 3:
Enter name of course 3: Computer Architecture
Enter credits for course 'Computer Architecture': 4
Enter Grade for course 'Computer Architecture': 3.94

Course 4:
Enter name of course 4: Engineering Project Management
Enter credits for course 'Engineering Project Management': 4
Enter Grade for course 'Engineering Project Management': 3.98

Course 5:
Enter name of course 5: Differential Equation
Enter credits for course 'Differential Equation': 4
Enter Grade for course 'Differential Equation': 3.97

=====--RESULT--=====
Student Name: Amima
Current Semester: 4

-----Course Details-----
```

```
=====--RESULT--=====
Student Name: Amima
Current Semester: 4

-----Course Details-----
Subjects:
Course Name: DSA
Credit Hours: 4
Grade: 3.95
Grade Points: 15.8

Course Name: Signals and Systems
Credit Hours: 4
Grade: 3.98
Grade Points: 15.92

Course Name: Computer Architecture
Credit Hours: 4
Grade: 3.94
Grade Points: 15.76

Course Name: Engineering Project Management
Credit Hours: 4
Grade: 3.98
Grade Points: 15.92

Course Name: Differential Equation
Credit Hours: 4
Grade: 3.97
Grade Points: 15.88

-----Current Semester GPA Details-----
Total Credits: 20
Total Grade Points: 79.28
Semester GPA: 3.964

-----CGPA Details-----
Enter GPA of Semester 1: 3.95
```

```
-----Current Semester GPA Details-----
Total Credits: 20
Total Grade Points: 79.28
Semester GPA: 3.964

-----CGPA Details-----
Enter GPA of Semester 1: 3.95
Enter GPA of Semester 2: 3.75
Enter GPA of Semester 3: 3.85
CGPA of Amima is: 3.8785
```

```
PS C:\Users\amima\OneDrive\Desktop\VS CODE folder> |
```

Testing:

The application was tested by performing each operation:

Sr.	Task	Result	Status
1.	Student information	Student name and current semester are accepted correctly.	Passed
2.	Course information	Course name, credit hours, and grade are accepted correctly.	Passed
3.	Total credits	Credit hours are calculated correctly.	Passed
4.	Total grade points	Grade points are calculated correctly.	Passed
5.	CGPA calculation	Overall CGPA is calculated and displayed correctly.	Passed
6.	Result display	Student and course details are displayed correctly.	Passed

Conclusion:

The CGPA Calculator successfully demonstrates the practical use of fundamental C++ and object-oriented programming concepts. The project helped in understanding classes, objects, constructors, vectors, member functions, loops, conditional statements, user input, references, and arithmetic operations.

The application provides a simple way to calculate semester GPA and overall CGPA while managing information about multiple courses.

GitHub Repository:

[AmimaBatoool/CodeAlpha CGPA-calculator: A simple C++ CGPA Calculator developed as part of my CodeAlpha C++ Programming Internship.](#)

Project status: Completed.